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Review

Core-shell systems of crosslinked organic polymers: A critical review

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ABSTRACT

Architecture of a crosslinked polymeric system is a critical point with respect to their applications. Core-shell is a modern and the most useful morphology of any system. In this regard, the classification, synthesis, characterization, and applications of core shell system of crosslinked polymeric network have been analytically reviewed in this article. In core–shell system, a crosslinked organic polymer is merged with another organic or inorganic material to form core–shell morphology. These core–shell systems of crosslinked polymers are used for drug delivery, adsorption of pollutants, anti-cancer, antibacterial, dye-sensitized solar cells, and catalysis as reported in literature. The applications of core–shell systems of crosslinked polymeric networks can be enhanced by adjusting the temperature, pH, glucose, and ionic strength which are discussed briefly. Conclusion along with future possibilities have also been emphasized to assist upcoming researchers focusing on core–shell morphology containing crosslinked polymers.

1. Introduction

The core–shell systems are composed of at least two distinct components from which one is present in core and other in shell region [1–3]. This morphology of materials has gained significant attention due to their amalgamation of superior properties which are not present in the individual components. These systems have the ability to combine the characteristics and behaviors of both the shell and core [4–6]. This morphology enables the translation of surface properties from the shell to the core [7,8]. In this way, these systems introduce new functionalities to these systems. Such microgels of crosslinked polymeric networks have various applications in various fields such as drug delivery [9], antibacterial activities [10], anticancer [11,12], catalysis [13], adsorption of pollutants [14], and dye-sensitized solar systems [15]. Typically, such systems are crafted into spherical shapes. They exhibit a particle structure in which the polymer initially polymerized is situated at the center of the particle, while subsequently formed polymers are integrated into the outer shell region [13]. From these systems, one or both regions of core–shell systems are made with soft material (crosslinked organic polymers) [16,17].

The crosslinked organic polymers exhibit the stimuli responsive behavior under various conditions of temperature [18], pH [19], and ionic strength [20]. Therefore, these factors control the efficiency of systems for various applications. The stimuli responsive behavior system decreases slightly with solid core (in solid core encapsulated with crosslinked organic shell) but more with solid shell (in crosslinked organic core encapsulated with solid shell) regions. Both core and shell regions can show stimuli responsive behavior if both (core and shell) regions consist of crosslinked organic (soft) materials [21,22]. In this way, the stimuli behavior of crosslinked organic polymers (soft materials) can be controlled as required for application such as control the leakage of loaded drugs [17] or metal nanoparticles [23], controlled

Abbreviations: AA, Acrylic acid; MAPTMS, 3-Methacryloxypropyltrimethoxy silane; AAm, Acrylamide; P(EI), Poly(ethyeleneimine); NIPMAM, *N*-isopropylmethacrylamide; NIPAM, *N*-isopropylacrylamide; MEEMA, 2(2-methoxyethoxy)ethyl methacrylate; NPs, Nanoparticles; P(NIPAM), Poly(N-isopropylacrylamide); P(EG), Poly(ethylene glycol); Si, Silica; MAA, Methacrylic acid; DMI, Dimethylitaconate; PT, Protein; P(S), Polystyrene; P(SR), Polysaccharide; CS, Chitosan; 4NPh, 4-Nitrophenol; CG, Collagen; P(GMA), Poly(glycidylmethacrylate; NVCL, *N*-vinylcaprolactam; NAMOBA, *N*-4-amino-2-methylene-4-oxobutanoic acid; HDD, Hydrodynamic diameter; VPTT, Volume phase transition temperature; HEMA, 2-Hydroxyethylmethacrylate; AAMPSAc, 2-Acrylamido-2-methylpropane sulphonic acid; MBAA, *N,N'*-methylenebisacrylamide; DLS, Dynamic light scattering; SDS, Sodium dodecyl sulfate; APS, Ammonium-per-sulfate; MEA, 2-Methoxyethylacrylate; UCST, Upper critical solution temperature; AET, 2-Aminoethylenethiol; AAEMA, Acetoacetoxyethylmethacrylate; DR, Doxorubicin; MC, Methyl cellulose; EDX, Energy dispersive X-rays; FTIR, Fourier Transformed Infrared Spectroscopy; GMA, Gelatin methacrylate; HRTEM, High resolution transmission electron microscopy; SEM, Scanning electron microscopy; SPR, surface plasmon resonance; TGA, Thermo gravimetric analysis; UV–Vis, UV/Visible spectroscopy; XPS, X-ray photoelectron spectroscopy; XRD, X-ray diffraction; DTT, Dithiothreitol; MMA, Methylmethacrylate; BSA, Bovine serum albumin; ICP-AES, Inductively coupled plasma-atomic emission spectroscopy.

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release of drug at target place [24,25], tunning the catalytic performance of hybrid systems [26], maximum adsorption of pollutants from water [14], and stability of metal nanoparticles in systems [27]. The stimuli responsive behavior of soft materials is due to the hydrophilic and hydrophobic interactions. These interactions play an important role in the swelling and deswelling behavior of soft materials. if the strength of hydrophilic interactions is greater than hydrophobic interactions then soft material is present in swelling state and if the strength of hydrophobic interactions is greater than hydrophilic than the soft material is present in deswelling state. The strength of these interactions depends upon the temperature and pH of the medium. For example, Suzuki and Kawaguchi [28] have synthesized core–shell system in which both core and shell regions are made with crosslinked organic polymers (soft materials) and studied the effect of temperature on swelling/deswelling behavior of these core shell systems. The core region is made with poly (glycidyl methacrylate) P(GMA) and shell with poly(*N*-isopropylacrylamide) P(NIPAM). Gold (Au) nanoparticles (NPs) were introduced into the shell region by *in situ* reduction method. Au nanoparticles are stabilized by these core–shell system and the hybrid core–shell systems showed swelling and deswelling behavior by varying the temperature value from 25 to 40 ℃. They reported more stability of gold nanoparticles in core shell system. They studied and reported the effect of temperature on gold nanoparticles loaded core shell system and color variation by showing the swelling and de-swelling nature of microgel simply by changing the temperature which can be monitored simply by naked eye due to variation in color. On the other hand, Chuang et al [29] have synthesized core–shell microgels in which core made with poly(*N*-isopropylacrylamide) and shell with *N*-4-amino-2-methylene-4-oxobutanoic acid (NAMOBA) and studied the effect of pH on swelling/deswelling behavior. The hydrodynamic diameter (HDD) of these core–shell systems decreased with increasing temperature (due to increasing hydrophobic interactions) and increased with increasing pH of the medium (due to conversion of –COOH groups into $–COO^-$ ions). They have used these core–shell systems for adsorption of copper (II) ions from water. The percentage removal efficiency of these systems is more at high pH values than low pH value.

According to scopus, 14,418 articles are reported on core–shell systems but out of these articles 495 articles are related to core–shell microgels. According to patent.google.com browser, A lot of patents are reported on core–shell systems. Therefore, different literature reviews on core–shell system [30–32] are reported but core–shell system consisting of crosslinked organic polymers with another organic, inorganic, and metal nanoparticles are not reported yet. Here in, a review is reported in which crosslinked organic polymers are present with other materials to form core–shell systems. The introduction of such systems has been reported briefly in section 1. In section 2, classifications of such systems on the basis of morphology basis have been discussed along with their suitability for applications. The synthetic approach and characterization techniques have been highlighted in section 3 and 4 respectively. The properties of core–shell systems under various conditions have been described in section 5. Various applications of core–shell systems have been explained in section 6. In the last, conclusion and future directions have also been given critically in section 7.

2. Classifications of core–shell microgels

The core–shell microgels can be divided into various classes on the basis of nature of materials used for their synthesis of core–shell microgels. These classifications are shown in Fig. 1 and their explanation is given below one by one.

2.1. Metal nanoparticles encapsulated crosslinked organic polymer

In these classes, the core region is made with metal nanoparticles which are then surrounded by organic crosslinked polymers as shown in Fig. 1(a). Such type of core–shell systems can be sued for catalytic reactions, but their catalytic performance is very low due to more distance between the reactants and the surface of metal nanoparticles [33]. Therefore, reactant molecules take more time in conversion into products. Such systems can be used as adsorbents for removal of different pollutants from environment. Polar parts are present in the structure of crosslinked networks which have the affinity to interact with pollutants. Therefore, pollutants can easily be adsorbed on the surface of these hybrid systems. Such systems can also be used for drug delivery due to stimuli responsive behavior of microgels. This responsive behavior is very fruitful in those systems in which Au nanoparticles are present in core region. These (Au) nanoparticles can increase the temperature by absorbing the short wavelength radiation. Therefore, the shell region of core–shell system is converted from swelling state to deswelling state [34–36]. Such systems can be further divided into (i) monometallic [37] and (ii) bimetallic core surrounded with crosslinked organic polymers. There bimetallic hybrid systems are better than monometallic due to high catalytic activity (due to synergistic effect) and easily reusable property (due to more density) [38].

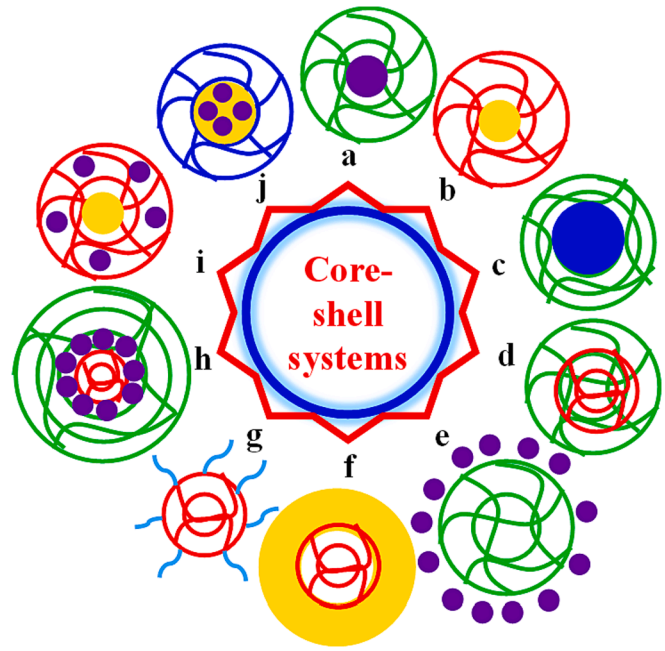

Fig. 1. Classifications of core–shell microgel systems (indication of color: Yellow for inorganic material; red, blue, green for organic materials; purple for metal nanoparticles) (**a**) metal nanoparticles encapsulated crosslinked organic polymer (**b**) inorganic particles encapsulated organic crosslinked polymer (**c**) organic polymer encapsulated crosslinked organic polymer (**d**) crosslinked organic polymer encapsulated another crosslinked organic polymer (**e**) crosslinked organic polymer encapsulated metal nanoparticles (**f**) crosslinked organic polymer encapsulated inorganic particles (**g**) crosslinked organic polymer encapsulated linear organic polymer (**h**) crosslinked organic polymer encapsulated with a layer of metal nanoparticles surrounded another crosslinked organic polymer (**i**) inorganic particles encapsulated crosslinked organic polymer decorated metal nanoparticles (**j**) metal nanoparticles decorated inorganic particles encapsulated crosslinked organic polymer. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

2.2. Inorganic core encapsulated organic polymers

In these classes, the core is made up with inorganic materials (SiO_2 or Fe_2O_3 or Fe_3O_4) which are surrounded by organic polymers as shown in Fig. 1(b) [14,39]. Such systems are very important due to easily property of recycling by simple centrifugation process or applying magnetic field [40]. These systems are excellent adsorbent because of their recycling and easy loading and releasing of pollutants from these systems [41].

Such systems are not very suitable for catalysis (very poor catalytic property) even after the introduction of metal nanopartilces in the core region as shown in Fig. 1(j). Their catalytic activity can be enhanced by introducing the metal nanoparticles in the shell region of core–shell systems as shown in Fig. 1(i). The stimuli responsive behavior of these systems is also decreased slightly due to solid core [42–44]. Such systems are also much suitable for drug delivery [40,45]. Another challenge for systems is to control the size of inorganic core. For this reason, the synthesis of such systems is not easy to control required sized nanoparticles.

2.3. Organic core encapsulated organic crosslinked shell

In such classes, the organic polymer core is encapsulated with another crosslinked polymer. Such class is further divided into further two classes as given below.

2.3.1. Solid organic polymer core encapsulated crosslinked organic polymer shell

In these classes of core–shell microgels, a solid organic polymer is encapsulated with crosslinked organic polymer as shown in Fig. 1(c). Generally, polystyrene is used as organic core in such systems which is encapsulated with another organic crosslinked polymer. Stimuli responsive behavior of such systems also decreases slightly due to solid organic core as reported in our previous review article [22]. Such systems can be easily recycled after their application. Such systems can be used as adsorbent due to the presence of crosslinked organic shell region [46]. The adsorption capacity of such systems depends on the polarity of the shell components [47]. These components also affect the selectivity of the adsorption of pollutants. Such systems can be used for drug delivery due to stimuli responsive behavior. Such behavior facilitates loading and releasing the drug from core-shell microgel systems. The size of polystyrene core can easily be controlled with the help of stabilizers. Therefore, such types of core shell systems can easily be synthesized as compared to inorganic core encapsulated organic polymers. Such systems cannot be used as catalysts due to the absence of metal nanoparticles. They are able for catalysis after the introduction of metal nanoparticles in these systems [27,48,49].

2.3.2. Crosslinked organic core encapsulated crosslinked organic polymer

In such types of core–shell microgels, the crosslinked organic core is encapsulated with another crosslinked polymer to form the core–shell system as shown in Fig. 1(d). Such systems are rarely reported but most suitable for drug delivery. In such systems, the drug can be loaded in organic core region which is protected from leaching by another organic shell [50,51]. The shell region is the most important in these systems because this region effects on the loading and releasing of drug with the help of stimuli responsive behavior [52,53]. In such systems, both core and shell regions show stimuli responsive behavior [54]. Generally, the crosslinked microgels which have high value of volume phase transition temperature (VPTT) are present in shell region and which have low value are present in core. Because, when the temperature reaches the value of VPTT of shell region, the drug already shifted from core to shell due to less value of VPTT of core materials. Such systems can be used for adsorption of pollutants from water. Such applications are not reported in literature with these systems. Such systems cannot be applicable for catalysis due to the absence of metal nanoparticles. They can be used as catalyst after the insertion of metal nanoparticles in these core–shell systems.

2.4. Crosslinked organic polymer encapsulated metal nanoparticles

In such types of core–shell microgels, the organic crosslinked polymer is covered with metal nanoparticles as shown in Fig. 1(e). Such systems are very important with respect to catalytic activity. The reactant molecules can easily be approached on the surface of metal nanoparticles which acts as catalysts [55,56]. Such systems are further divided into two classes on the basic of monometallic [57] and bimetallic [58] nanoparticles. But the catalytic activity of those systems in crosslinked polymer is surrounded with bimetallic nanoparticles is higher than monometallic systems due to synergistic effect. Such systems can also be used as adsorbent, but their adsorption capacity is lower than previously discussed classes. The loading of pollutants is slightly restricted by metal nanoparticles. The metal nanoparticles also effect on the loading and releasing of drug from outside to inside of microgels and inside to outside. In this way, such systems are not very suitable for drug delivery. Therefore, such applications of crosslinked organic polymer microgels encapsulated metal nanoparticles are not much reported in literature.

2.5. Crosslinked organic polymer encapsulated with inorganic materials

In such classes, the crosslinked organic polymer is surrounded by inorganic (SiO_2 or Fe_2O_3 or Fe_3O_4) materials as shown in Fig. 1(f). In such systems, the stimuli responsive behavior of smart microgels is controlled by the thickness of inorganic shell. If the thickness of inorganic materials is high, then the microgel core shows little responsive behavior and if the thickness of inorganic shell is low then the smart microgels show more responsive behavior [59,60]. But the stimuli responsive behavior of smart microgels are greater than these systems. The shell region of inorganic materials helps to control the leaking of loaded drug (during drug delivery) and pollutant (during recycling by centrifugation). Such systems can be used for drug delivery and biosensors [61]. The density of such systems is very high due to inorganic materials. Therefore, they can easily be recycled by simple centrifugation. Such systems can also be used for adsorption of different pollutants. Such systems are not suitable for catalysis, but they can be made able for catalysis after integration of metal nanoparticles in these core–shell systems. Such systems can also be used for adsorption of water from humid areas.

2.6. Organic polymer encapsulated linear organic polymer

In such classes, the crosslinked organic polymer is surrounded with linear organic polymers as shown in Fig. 1(g). Crosslinking is present in only the core region, but this is not present in the shell region. Such systems have more adsorption capacity than other core shell systems due to large active sites. Such systems can adsorb the pollutants from the surface of organic core as well as the linear polymer. Therefore, large number of pollutants can be adsorbed in such core–shell systems [52,62]. Such systems do not show catalytic activity, but they can be used as catalyst after integrating the metal nanoparticles in these core–shell systems [63]. Such types of core shell microgels are not easily recycled. During recycling by centrifugation, large amount of metal nanoparticles detached (leaked) from such core–shell system. Such systems also be used for drug delivery but the control of leakage of drug from these systems is not easy. Due to these drawbacks, a few articles are present in such core–shell systems. To control the nanoparticles or loaded drug inside these core–shell systems is a challenge for future research.

2.7. Organic core trapped a layer of metal nanoparticles encapsulated organic polymers

In such classes, organic polymer is surrounded with a layer of metal nanoparticles and then organic polymers shell as shown in Fig. 1(h). Such systems have more catalytic activity than metal nanoparticles encapsulated microgel systems but lesser form those in which the metal nanoparticles are present in shell region or on the surface of microgels. The catalytic activity of such systems can be enhanced further by integration of metal nanoparticles in shell region of these systems [64]. Such systems can be easily recycled due to the high density of core shell

systems. Such systems can be used for adsorption, but no such application has been reported yet. These are vacant spaces in the research field for the future.

3. Synthesis

The core–shell microgel systems can be synthesized by the following methods.

3.1. Single step synthesis of core–shell microgels

In this method, the monomers and crosslinkers are mixed simultaneously and heated at a specific temperature and pH of the medium with constant stirring. Of these monomers, one monomer is more polar and the other is less or non-polar in nature. The non-polar and less polar monomers have less affinity towards water molecules while highly polar monomers have high affinity towards water molecules. The non-polar and less polar monomers are hydrophobic in nature and polar monomers are hydrophilic in nature. When these monomers are mixed simultaneously in water in the presence of crosslinker at a specific temperature and pH, the hydrophobic monomers start to coagulate rapidly due to hydrophobic interactions. While the hydrophilic monomers start to coagulate slowly due to interactions with water molecules. Therefore, non-polar and less polar monomers make core and shell regions are made with polar monomers. Stabilizer is also used in this method to control the size of core–shell system. Mustafa et al [13] have also reported core–shell microgel system in a single step as shown in Fig. 2. They added *N*-isopropylacrylamide (NIPAM) (monomer), 2-hydroxyethylmethacrylate (HEMA) (monomer), *N,N′*-methylene-bis-acrylamide (MBAA) (crosslinker), and sodium-dodecyl-sulfate (SDS) (stabilizer) in a water containing tree neck flask. The mixture stirred under nitrogen and then the temperature raised till 75 ℃. The reaction continued for 30 min. Then a small amount of ammonium-per-sulfate (APS) was added dropwise to start polymerization and reaction continued for further 5 h under similar conditions. The appearance of milky color indicated the formation of microgel. The product was dialyzed for purification. In this system, the core is made with HEMA monomer and shell with NIPAM monomer. Melle et al [52] have also synthesized core–shell microgels with using *N*-vinylcaprolactam (NVCL) (monomer), 2-methoxyethyl acrylate (MEA) (monomer), and *N, N′*-methylene-bis-acrylamide (MBAA) (crosslinker) in a single step. In this system, NVCL is present in crosslinked core region and MEA as linear polymer shell. Zeng et at [65] have also reported the synthesis of core–shell system in a single step. In this system, polystyrene P(S) is present in the core region and poly(*N*-isopropylacrylamide-acrylic acid) P(NIPAM-AA) in shell region.

In this method, less time is required for synthesis of core–shell microgel systems. But the size control of core–shell systems is not an easy task in this method. The crosslinking density and compositions of core–shell systems cannot be controlled properly in this method. These are the challenges for future research.

3.2. Two step synthesis of core–shell microgels

In this method, the core is synthesized first and shell region in the second step. All types of core–shell microgel systems can be synthesized by this method. Generally, less polar, or non-polar core is synthesized first, and their size also controlled with the help of stabilizer at a specific temperature and controlled addition of free radical initiator. The size controlled of core by these parameters is beyond the scope of this review article. Therefore, these parameters are not discussed here. During the formation of the core, some amount of polar part is also added in this core to it able for interaction with the polar shell region. The shell region is mostly made by polar monomers with and without comonomers. The thickness of shell region depends on the percentage of monomers and comonomers used during the synthesis of shell region around core. Arif et al [27] have synthesized polystyrene P(S) encapsulated poly(*N*-isopropylmethacrylamide) P(NIPMAM) core–shell microgels in two steps as shown in Fig. 3. In the initial phase, styrene (S) (monomer), NIPMAM (polarity inducer), and SDS (stabilizer) were added into deionized water containing a three-neck flask. The mixture stirred for 10 min at 250 rpm in N_2 supply. The temperature was then elevated and adjusted at 80 °C for 30 min. Subsequently, a small amount of ammonium per sulfate solution was introduced into the flask, and the reaction proceeded for 7 h under N_2 supply. The resulting polystyrene core latex dispersion underwent a 4-day dialysis to eliminate any remaining reactants. In the subsequent stage, the P(S) core latex particles were utilized as seeds to generate a poly(*N*-isopropylmethacrylamide) P(NIPMAM) shell around them to form core–shell microgels. For this synthesis, NIPMAM, P(S) latex dispersion, *N, N′*-methylenbisacrylamide (MBAA) were put into a deionized water containing round-bottom flask and stirred at 250 rpm for 5 min under N_2 supply. The temperature gradually increased to 70 °C at a rate of 1 °C per minute and maintained under N_2 supply. After 30 min of temperature stabilization, a small amount of ammonium-per-sulfate (APS) was added to the flask and stirring at the same rate continued for an additional 4 h to complete the reaction. The resulting P(S)@P(NIPMAM) core–shell microgel dispersion dialyzed for 3 days to remove any unreacted surfactant, components, and initiator. Cao-Luu et al [59] have also reported the synthesis of core–shell system in which organic core is encapsulated with silica shell. Wei et al [66] have also synthesized Au nanoparticles encapsulated poly(N-isopropylacrylamide) hybrid core–shell microgels in two steps.

This method is very important and easy to control the size of core and thickness of shell region. These two factors are very important with respect to their applications. All types of core–shell systems can be synthesized by this method.

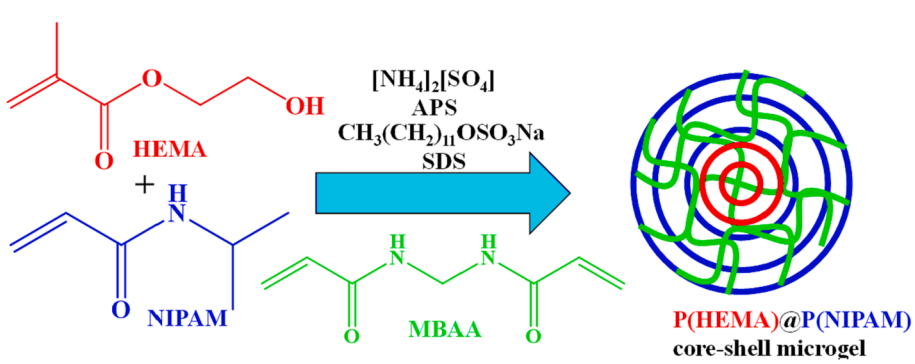

Fig. 2. Pictorial diagram of single step synthesis of poly(2-hydroxyethylmethacrylate)@P(N-isopropylacrylamide) P(HEMA)@P(NIPAM) core–shell microgels [13].

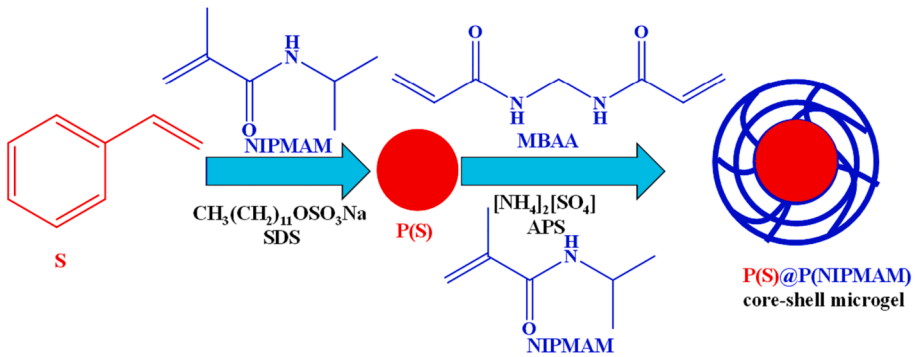

Fig. 3. Pictorial diagram of two step synthesis of polystyrene@poly(*N*-isopropylmethacrylamide) P(S)@P(NIPMAM) core–shell microgels.

3.3. Simultaneous mixing of separately synthesized core and shell materials for core–shell microgel formation

In this method, the core and shell materials are first synthesized separately and then both components (component of core formation and components of shell formation) are mixed simultaneously. For such systems, the core and shell formatting materials are first synthesized and then stirred for formation of core–shell systems. When both components are mixed and stirred then electrostatic interactions take place. Due to these electrostatic interactions, the shell components come on the surface of core and start to coagulate in the form of shell. The thickness of this shell region is controlled by controlling the coagulation of shell components with the help of temperature, concentration of shell components, rate of stirring and pH of the medium. Generally, microgels encapsulated metal nanoparticles core–shell hybrid systems are synthesized in this method. Mutharani et al [57] have also synthesized crosslinked polymer microgels encapsulated palladium nanoparticles hybrid core–shell microgel systems as shown in Fig. 4. They synthesized microgels and palladium nanoparticles separately and mixed them in a beaker while stirring. The mixture was stirred for 1 h at 200 rpm. Palladium nanoparticles interact with crosslinked organic polymer microgel on mixing and the stirring process facilitates the chance of collisions between metal nanoparticles and crosslinked organic polymer. In this way, the interactions between microgels and metal nanoparticles reach their maximum level. Gu et al [67] also have reported the synthesis of core–shell hybrid microgel systems by mixing of microgels and metal nanoparticles.

In this method, microgels encapsulated metal nanoparticles core–-shell hybrid systems have been reported in literature. No synthesis of other systems is reported in literature yet. It is a challenge to synthesis other core–shell systems with this method. The stability of these synthesized hybrid systems is not too much because metal nanoparticles, which are present on the surface of microgels, start to coagulate after a few days.

4. Characterization techniques

Various techniques are utilized to explore the characteristics of core–shell systems of microgels. These characterizations are encompassed on their behavior, size, composition, structure, and appearance. These approaches also serve to differentiate the components of core and shell regions of core–shell microgels. The specific techniques used for characterization are: laser light scattering spectrometry (LLS), dynamic light scattering (DLS), wide-angle X-ray scattering (WAXS), scanning electron microscopy (SEM), photoluminescence spectroscopy (PL), 1H-nuclear magnetic resonance spectroscopy (NMR), UV–Visible spectrometer (UV–Vis), Raman spectroscopy (RS), transmission electron microscopy (TEM), energy-dispersive X-ray spectroscopy (EDX), atomic force microscopy (AFM), photo correlation spectroscopy (PSC), fourier transform infrared spectroscopy (FTIR), attenuated total reflectance spectroscopy (ATR), differential mechanical analysis (DMA), polarizing optical microscopy (POM), differential scanning calorimetry (DSC), and

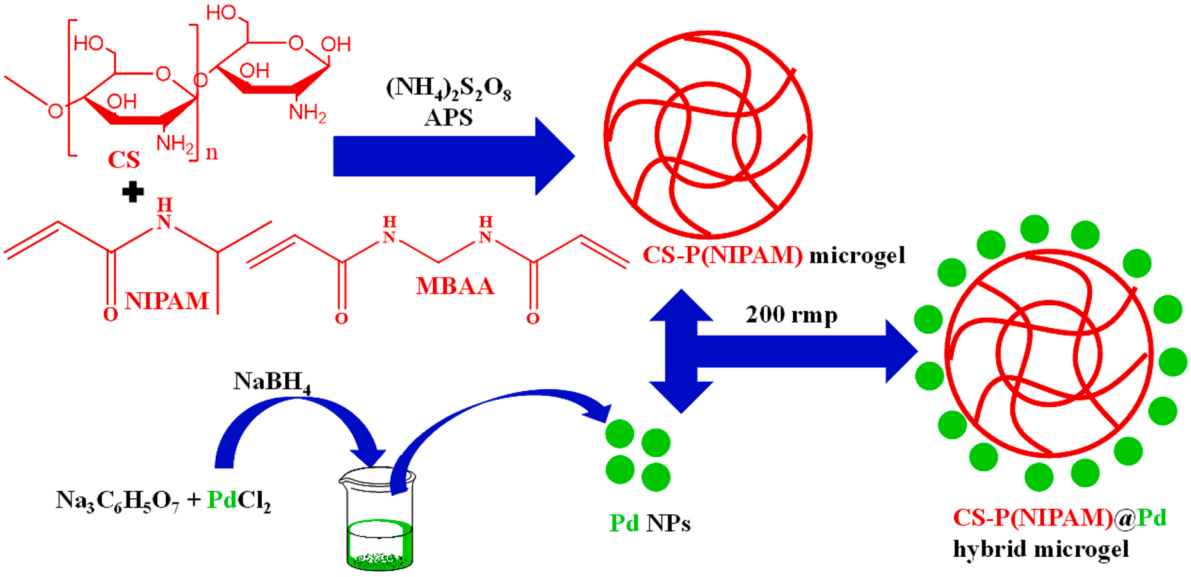

Fig. 4. Pictorial diagram of synthesis of core–shell hybrid system by mixing of Pd nanoparticles with chitosan-poly(N-isopropymacrylamide) CH-P(NIPAM) microgels [57].

X-ray diffraction (XRD).

The techniques such as inductively coupled plasma mass spectrometry (ICP-MS), inductively coupled plasma atomic emission spectroscopy (ICP-AES), inductively coupled plasma-optical emission spectrometry (ICP-OES), and brunauer emmett teller (BET) are employed to determine the elemental compositions in biological fields and pore size and surface area of microgels respectively.

RS, NMR, and FTIR techniques are extensively employed to investigate the functionalities of resultant organic polymers and to discern the interactions within the polymeric network of microgels combined with other components in core–shell systems. XRD and EDX techniques serve to validate the presence and characteristics of metal nanoparticles within core–shell systems. XRD is specifically utilized to determine the crystallinity of core–shell hybrid microgels. WAXS, Particle Size Analyzer, and DLS are utilized for measuring the diameter and size of core–shell microgel systems. UV/Vis/NIR spectroscopy is applied to ascertain the volume phase transition temperature (VPTT) of core–shell microgel systems by analyzing turbidity measurements. Moreover, it can be employed to verify the stabilization and creation of plasmonic metal nanoparticles embedded in microgels. It also helps to determine the loading and releasing of drug in core–shell systems.

Advanced microscopic instruments such as TEM, AFM, and SEM are employed to analyze the physical attributes of microgel particles. TEM is used to investigate the morphology, size, and size distribution of core–shell microgels along with dimensions and shapes of polymer microgels. Meanwhile, AFM and SEM are utilized to examine the surface morphology of core–shell organic polymers.

The examination of drug release entails evaluating the cytotoxicity of microgel and hybrid microgel systems through MTTS analyses. Quantification of the released drug is performed using the HPLC technique, while the monitoring of drug loading and release is observed via confocal laser scanning microscopy (CLSM). ICP-MS, ICP-OES, and ICP-AES are utilized to determine the quantity of noble metal nanoparticles integrated into microgels.

DMA, DSC, and TGA techniques are utilized to explore the decomposition and thermal stability characteristics of core–shell microgel systems. TGA is additionally employed to quantify the percentage of core components and the percentage of shell components of core–shell systems. DLS/Photon correlation spectroscopy (PCS) is employed to determine the size distribution and hydrodynamic size of both core and core–shell microgels. WAXS is used to recognize the structure and consistency of synthesized hydrogels, providing valuable insights into the network topology and ensuring reproducibility of structures. Rheological measurements are carried out to understand the viscoelastic properties of hydrogels. Table 1 consolidates information on the core–shell microgels with core, shell, the characterization methods employed, and their respective applications.

5. Properties of core–shell systems

Smart crosslinked polymeric network of core–shell microgel systems exhibit response under changing the stimuli conditions, such as temperature, ionic strength, pH, and light as shown in Fig. 5. These stimuli conditions produce sudden changes in the hydrodynamic diameter of core–shell system, which are dependent on their specific compositions. These stimuli conditions will be discussed in these sections.

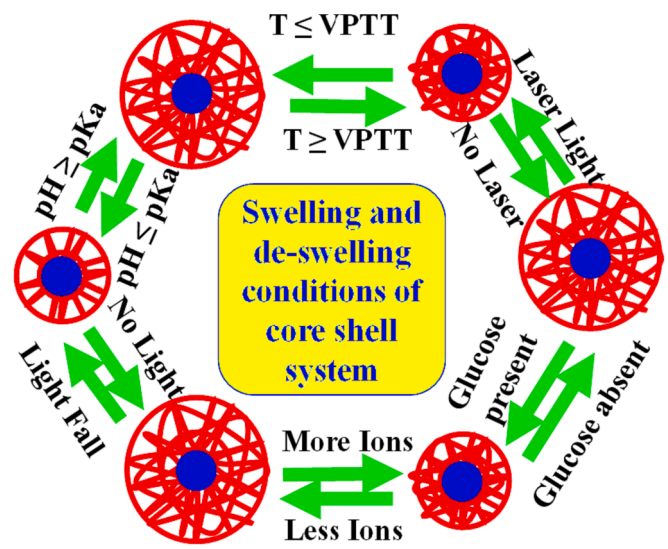

Fig. 5. Swelling/deswelling behavior of core–shell systems under different conditions of pH, ionic strength, light, temperature, and glucose.

Table 1
Systems of core–shell microgels along with core and shell materials, applications of core–shell systems and the used characterization techniques for their analysis.

System	Core	Shell	Applications	Characterization Techniques	References
Si@P(NIPAM-DMI), @P(NIPAM-DMI)	Si	NIPAM, DMI	Loading of protein	DLS, Zeta-sizer, UV–Vis, TEM, FLIM	[68]
PT@P(SR)	PT	SR	Stabilizer	UV–Vis, TEM, SEM,	[69]
Si@P(NIPAM-MAA)	Si	NIPAM, MAA	Drug deliver	SEM, DLS, TEM, FTIR, MTT, UV–Vis, TEM	[70]
P(S)@P(NIPAM)	S	NIPAM	Not used	SEM, FESEM, DLS, Zeta-sizer	[71]
P(S)@P(MMA)	S	MMA	Adsorption of CO_2	^{1}H NMR, ^{13}C NMR, SEM, TEM, TEM, DLS, DSC, DMA	[72]
CS@CG	CS	CG	Drug deliver	SEM, FTIR, TGA, TG, Particle sizer	[73]
P(S)@Au-P(NIPMAM)	S	NIPMAM	Catalysis	UV–Vis, TEM, FTIR	[27]
P(S)@Ag-P(NIPMAM-AA)	S	NIPMAM, AA	Catalysis	DLS, TEM, UV–Vis, FTIR	[74]
P(NIPAM)@P(NIPAM-AA)	NIPAM	NIPAM, AA	Not used	DLS, FTIR, AFM, OM, SEM, STEM, TEM	[75]
P(MMA)@P(EG)	MMA	EG	Drug delivery	SEM, DLS, TEM	[17]
Au@P(NIPAM)	Au	NIPAM		DLS, AFM, SEM, *Optical spectroscopy*	[76]
P(NIPMAM)@P(NNPA), P(NNPA)@P (NIPMAM)	NIPMAM	NNPA	Sensor	FTIR, PCS,	[77]
MC@GMA	MC	GMA	Drug delivery	BFI, CLSM, SEM, *particle sizer*	[78]
Au@Ag@P(NIPAM)	Au@Ag	NIPAM	Sensor	UV–Vis, FE-SEM, ATR-FTIR, AFM	[79]
P(S)@Ag-P(NIPMAM-AA)	S	Ag, NIPMAM	Catalysis	TEM, FTIR, UV–Vis., DLS,	[80]
Fe@Pd-P(NIPAM-MMA)	Fe	Pd, NIPAM MMA	Catalysis	ED-XRD, SEM, ICP-OES, FIB-SEM, DLS, EDXS,	[81]
CS@P(EG)	CS	EG	Drug delivery	DLS, SEM, *Particle sizer*, TEM, FTIR	[82]
P(S)@P(NIPAM)	S	NIPAM	Catalysis, adsorption	TEM, UV–Vis., XRD, DLS, FTIR	[46]
Fe_3O_4@P(NIPAM)@CS	Fe_3O_4	NIPAM, CS	Drug delivery, Anticancer	FTIR, VSM, TGA, FE-SEM, EDX, DLS, TEM	[83]
P(S)@P(NIPAM)	S	NIPAM	Not used	SEM, FTIR, UV–Vis.,	[84]

5.1. Temperature sensitivity

Smart microgels show swelling and deswelling behavior under various conditions of temperature. The crosslinked polymer microgels have polar groups in their structures such as –COOH, $–NH_2$, $-SO_3H$, –NHCO- etc. These polar groups make dipole–dipole interaction and hydrogen bonding with water molecules. Therefore, water molecules can come inside the crosslinked polymeric network due to this force of interactions. On the other hand, dipole–dipole interactions and hydrogen bonding are also present in the structure of crosslinked networks. The force of interaction due to which water molecules come into the structure of microgels is called hydrophilic and the interactions due to which the water molecules come out the crosslinked network is called hydrophobic interactions. When the temperature of the medium increases, the hydrophilic interaction starts to decrease, and hydrophobic interactions begin to become a strong force of interaction. Therefore, hydrodynamic diameter (HDD) of microgels decreases with increasing the temperature. But, at a specific temperature, the hydrodynamic diameter of microgels decreases rapidly in classes volume phase transition temperature (VPTT). The value of VPTT has a specific value for a specific type of microgels. While some core–shell systems show opposite behavior under various conditions of temperature. The HDD of core–-shell systems can be increased with increasing temperature [85]. The temperature at which the HDD of such core–shell systems increase rapidly is called upper critical solution temperature (UCST). After this temperature, the volume of core shell system increases rapidly on further increasing the temperature. This value can be altered by adding comonomers [86] and varying the crosslinking density [87] in microgels [88].

The variation in HDD of microgels is also greatly affected by their position in core–shell systems and the nature of other materials [89,90]. If the microgels are encapsulated with inorganic materials such as SiO_2, the effect of temperature is significantly affected on the thickness of SiO_2. If the thickness of SiO_2 shell is high then the core–shell system is less sensitive towards heat, while if the thickness is small resulting more sensitive core–shell system (there prominent effect is still less as pure microgels) [59]. The systems in which the crosslinked network is present in shell region have more prominent temperature sensitivity (there significant effect is also less than pure microgels). Suzuki and his coworkers [28] have synthesized core–shell systems in which poly(glycidyl methacrylate) P(GMA) is present in core region and poly(*N*-isopropylacrylamide) P(NIPAM) in shell region. The shell region was then modified with 2-Aminoethanethiol (AET) or dithiothreitol (DTT) or both of these components. The structures of these core–shell systems have amide, amine, ester, –OH and thiol groups. These groups can interact with water molecules in the following order.

Thiol group < Amide group < Amine group

They reported that when the amount of NIPAM increased in shell region then the swelling behavior of core–shell system increased because more amide groups were introduced into the shell region which interact strongly with water molecules and HDD of core–shell systems increases. When the DTT has introduced into shell region then their HDD also increased but not as increased in the former system because the pH of medium was slightly acidic (pH = 6) and DTT have the ability to make cyclic ring to some extent and thus decreased the interactions towards water molecules and the structure of core–shell. Therefore, these systems have less affinity towards water molecules. When AET was presented into the shell then HDD increased more as compared to NIPAM because $–NH_2$ groups make hydrogen bonding with water molecules and hence have larger affinity towards water. Liu et al [91] have also reported similar effect of temperature on a core–shell hybrid microgels. Schmid et al [92] illustrated that adjusting the temperature enables the regulation of the permeability of inner shell, with the outer shell retaining its swollen state and ensuring the colloidal stability of the particle. The thickness of shell can be controlled by hydrophobic interactions and the inversion of core and shell can also be possible [93]. The temperature responsive behavior of core–shell microgels are also be affected by varying the content of constituents of core–shell systems to produce crystallinity [94] and fluid interface of monolayer core–shell systems [95–97].

5.2. pH sensitivity

The core–shell microgels have acidic or basic characteristics on the basic of acidic (–COOH, $-SO_3H$) or basic ($–NH_2$) groups. When these groups are present in the structure of core–shell microgels, then the core–shell systems have the affinity to donate or accept the protons to form ionic structures. Therefore, electrostatic interactions take place between the structures of core–shell systems. These interactions alter the value of HDD of core–shell microgels.

The core–shell microgels, which have –COOH [98–100] or $-SO_3H$ [101,102] groups in their structures, are called acidic core–shell microgel system while if the structure of core–shell systems contains $–NH_2$ groups are called basic core–shell systems. The acidic groups of acidic core–shell systems donate their protons from acidic groups (–COOH, $-SO_3H$) of their structures at pH ≥ pKa. After donation of their protons, the carboxylic groups are converted into carboxylate ($–COO^-$) ions. Due to this conversion, electrostatic repulsion takes place between the structures of core–shell systems. Therefore, the HDD of acidic core–shell systems increases after pH ≥ pKa. On the other hand, basic core–shell systems ($–NH_2$ containing core–shell systems) [103] can accept the protons at pH ≤ pKa. After accepting the protons, the amino groups of basic core–shell systems are converted into ammonium ($–NH_3^+$) ions. After this conversion, electrostatic repulsion takes place between the structures of core–shell systems. Due to this repulsion, the HDD of core–shell systems increases from normal value. This swelling and deswelling behavior of acidic and basic core–shell systems are reversible. Agrawal et al [104] have also reported the synthesis of a core shell system in which core consist of strongly crosslinked poly(acetoacetoxyethylmethacrylate) P(AAEMA) and shell of weakly crosslinked poly(*N*-vinylcaprolactam-acrylic acid) P(NVCL-AA). The HDD of the core–shell system increased rapidly by increasing the pH of the medium at pH ≥ pKa. Such behavior was achieved due to the releasing of protons from carboxylic acid groups of polymeric networks to form carboxylates ($–COO^-$). Due to this conversion, repulsions took place between the carboxylate ions and hydrodynamic diameter increased. At pH ≤ pKa, the acidic carboxylic groups are present in protonated forms (–COOH) and there is no electrostatic repulsion between the structures of core–-shell microgels. Therefore, the hydrodynamic diameter decreased again. Such behavior becomes more prominent at higher concentration of comonomer (AA) due to more carboxylic groups in the structures of core–shell microgel systems. Therefore, stronger electrostatic repulsion takes place as compared to lower content of AA in core–shell systems. Zuo et al [105] have synthesized core–shell system as poly(*N*-isopropylacrylamide-3-methacryloxypropyltrimethoxysilane) P(NIPAM-MAPTMS) core and poly(acrylic acid) P(AA) shell. Au nanoparticles were then generated in the shell region of these core–shell systems. The presence of carboxylic groups in shell region is more effectively affected by changing the pH of the medium. The effect of pH on hybrid core–shell system was identified by using UV–Vis. spectrophotometry. When the pH increased from 5 to 7, the value of HDD of core–shell systems increased rapidly due to deprotonation of carboxylic groups. Leung et al [106] have synthesized core–shell system in which P(NIPAM) was present as core and poly(ethyleneimine) P(EI) as shell. They examined the effect of pH on swelling behavior of core–shell system by varying from 3 to 7. The hydrodynamic diameter of core–shell systems decreased increasing the pH of medium due to deprotonation of ammonium ions (conversion from $–NH_3^+$ to $–NH_2$ form). Therefore, electrostatic repulsion decreases due to this conversion and hence, the size of core–shell system decreases.

5.3. Light sensitivity

The light or photons are basically the energies and these energies also effected on the swelling and deswelling behavior of core–shell systems [35]. When light or photons are fallen on the core–shell microgels, the hydrophobic interactions increase, and hydrophilic interactions decrease. Therefore, the hydrodynamic diameter of core–shell systems decreases with increasing the intensity of light or photons. Due to this property, core–shell systems can be used for drug delivery. The drug is loaded into the core–shell systems at low temperature and injected into the body. Laser light or photons fall on the drug loaded core–shell system when the drug loaded system reaches the target place. Due to this light, the core–shell system shrinks, and the water molecules come out from the core–shell system along with loaded drug. But this effect is not more significant. This effect can be made more prominent in the presence of Au nanoparticles [107]. The presence of Au nanoparticles also increases this deswelling behavior of hybrid core–shell systems. Au nanoparticles have the capacity to convert light into heat. Therefore, deswelling occurs when applying laser light or photons on core–shell systems. Budhlall and his coworkers [108] have synthesized Au NPs core and poly(*N*-isopropylacrylamide-acrylamide) P(NIPAM-AAm)) shell of core–shell system. They studied the effects of light, temperature, and microwaves on these hybrid core–shell systems. More prominent deswelling behavior is observed in hybrid core–shell systems instead of core–shell systems alone. Similar observations are reported in Au-based core–shell systems by Khaletskaya et al [109].

5.4. Ionic strength sensitivity

The amount and nature of metal salts also affect the swelling/deswelling behavior of core–shell systems. The metallic ions have positive charge, and the structure of core–shell systems have polar components. These components interact with the metal cations in the form of ion–dipole interaction or ion-ion (electrostatic interaction) interactions. The electrostatic (ion-ion) interaction is stronger than ion–dipole interactions. The acidic and basic core–shell systems have the affinity to donate and accept the protons from their structure respectively. When the acidic core–shell systems donate the protons from their structure to form anions. These anions strongly interact with metal cations like Na^{+} ions due to electrostatic interaction. Therefore, the anionic components come close to each other due to this electrostatic interaction. This interaction can be increased by increasing the content of metal salts. In this way, the content of metal salts also affects the value of VPTT of core–shell systems. The value of VPTT of core–shell systems decreases in the presence of metal salts like NaCl. These metal salts concentration also affect the aggregation of metal nanoparticles. At high content of NaCl, the HDD of core–shell systems decreases. Therefore, metal nanoparticles come closer to each other and start to coagulate to a specific level. Strozyk et al [110] have synthesized Au NPs based core–shell system in which Au NPs are present as core and both bovine serum albumin (BSA) and 2-(2-methoxyethoxy)ethyl methacrylate (MEEMA) as shell regions. The value of VPTT decreased in the presence of NaCl due to decreasing the hydrodynamic diameter of core–shell system. Therefore, the Au nanoparticles aggregated to some extent and the size of Au NPs increased. The aggregation effect is not prominent at high pH value (pH = 10) than lower (pH = 2.5). At pH = 2.5, VPTT value was 40.5 ℃ in the absence of NaCl while 36.5 ℃ in the presence of NaCl (150 mM of NaCl). At pH = 10, No aggregation occurred in the absence of NaCl, while aggregation happened at 37–38 ℃ in the presence of sodium (150 mM of NaCl) salt.

Zuo et al [105] have synthesized Au NPs loaded into shell region of poly(*N*-isopropylacrylamide-3-methacryloxypropyltrimethoxysilane) @poly(acrylic acid) P(NIPAM-MAPTMS)@P(AA) core–shell system and investigated the effect of concentrations of copper ions. Cu^{+2} ions have stronger interaction with carboxylic groups of these core–shell systems due to stronger ion–dipole interaction. These ion–dipole interactions are stronger than hydrogen bonding which is present between the core shell systems and water molecules. Therefore, the water molecules come out from the core–shell systems. Therefore, the hydrodynamic diameter of hybrid core–shell system decreases with increasing the content of copper (II) ions. This effect is observed in both acidic and basic conditions. But such behavior is more prominent in basic conditions because in basic conditions, the carboxylic groups of core–shell systems are converted into carboxylate anions. Conversely, stronger electrostatic interactions took place at high pH value due to oppositely charged species (carboxylate containing core–shell microgels and Cu^{+2} ions). Therefore, more deswelling occurs on increasing the content of Cu^{+2} ions.

5.5. Glucose sensitivity

Phenyl-boronic acid-based core–shell microgel systems are designed specifically to detect the presence of glucose and precise quantification of glucose levels in blood samples. The covalently bonded –OH groups of boronic acid react with the OH groups of glucose to form an ether linkage during the detection process and it is reversible as shown in Fig. 6 [111].

In the absence of glucose, the phenyl-boronic acid containing core–shell microgel structure exists in a dynamic form between the charged and uncharged states in water medium. However, upon the introduction of glucose, there is a shift in this equilibrium position to form a charged core–shell microgel complex derived from phenyl-boronic acid-based core–shell systems [112]. Consequently, the core–shell systems exhibit an increased charged composition as the glucose concentration rises [113]. Due to these charged compositions, the hydrophobic interactions between the crosslinked network of core–shell system also increases. Therefore, the hydrodynamic diameter of core–shell systems decreases [114]. If these interactions are present in the shell region, the diameter of core region is also affected with the deswelling and swelling behavior of shell regions [115]. The glucose responsive core–shell systems can be used for releasing of insulin as reported by Bai et al [116].

6. Applications of core shell microgel systems

Core-shell microgel systems have various applications in different fields as shown in Fig. 7. These applications are discussed below one by one.

6.1. Biological applications

Core-shell microgel systems can be used in biological applications due to their stimuli responsive behavior [117] and carrier [118] properties. Their biological applications are given below.

6.1.1. Drug delivery

Core-shell systems are used for drug delivery to the target place. The stimuli responsive behavior of core–shell systems makes these systems suitable for drug delivery [59,119–121]. Due this this property, the drug can easily be loaded and released at the target place [122–124]. Basically, the core–shell systems act as carrier for drug. Generally, the diameter of core–shell systems is greater at lower temperature due to swelling state [125,126]. In this state, more amount of drug is loaded into the core–shell systems and then these drugs loaded systems are injected into the human body. When the drug loaded systems reach the target place, photons of laser light are then fallen on these loaded systems. The systems shift from swelling state to deswelling state. During this conversion, the water molecules come out from the sieves of core–shell systems and the loaded drug also come out from the systems along with water molecules [127]. Such core–shell systems can also be used for treatment of cancer [128–130]. The thermo-responsive behavior becomes more prominent if the core–shell systems consist of Au nanoparticles in their structures. Such systems are more effective for treatment of cancer due to conversion of light into heat by Au

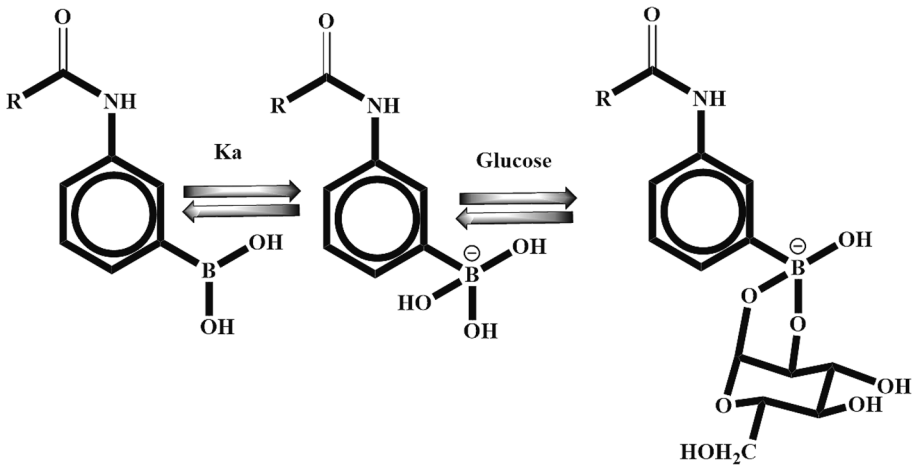

Fig. 6. Glucose sensitivity reaction mechanism by core–shell microgels [111].

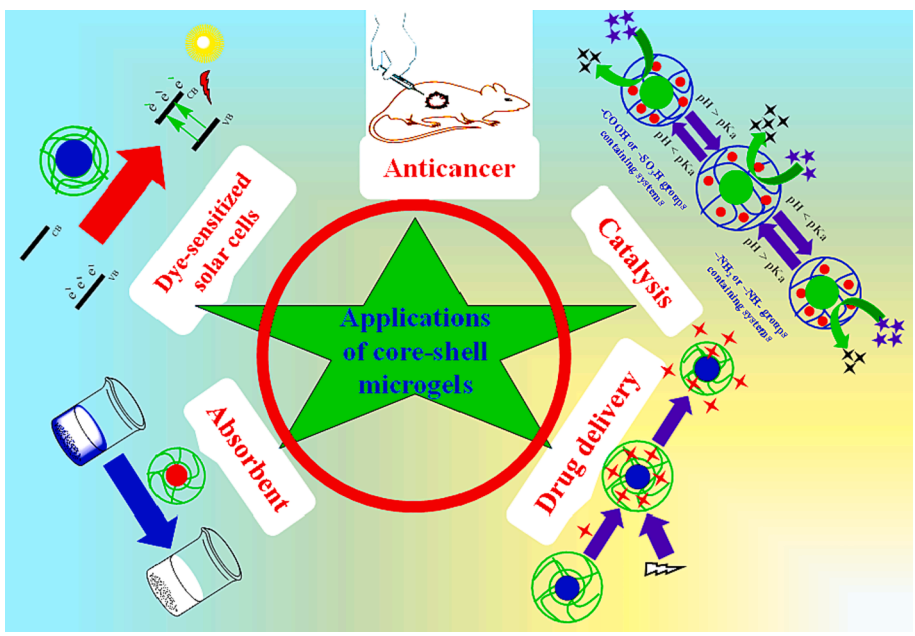

Fig. 7. Applications of core–shell microgel systems.

nanoparticles [131,132]. Nakamura et al [133] have synthesized Au nanoparticles containing core–shell hybrid systems. They investigated the effect of temperature and pH on swelling behavior of these hybrid systems because these factors affect the loading and releasing of the drug. Ghorbani et al [131] have also synthesized another core–shell system which contains Au nanoparticles. The loading and releasing effect of doxorubicin in these hybrid core–shell systems was investigated at various pH and temperature. was loaded on this hybrid core shell system. The drug released rapidly at pH = 5.3 due to losing the electrostatic attractions and released slowly at pH = 7.4 (due to the presence

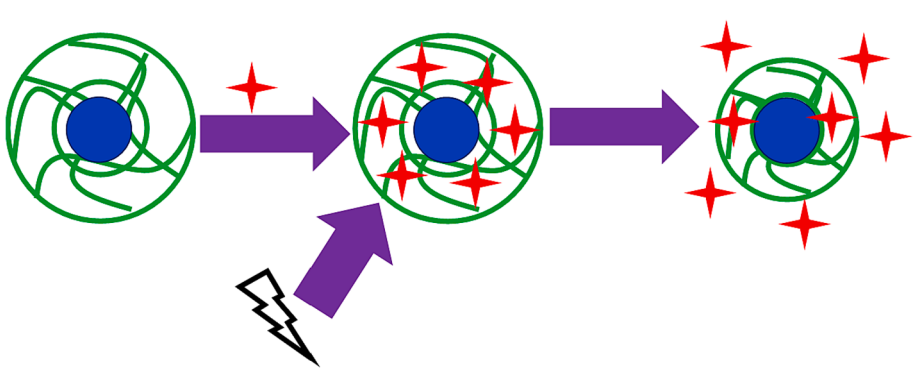

Fig. 8. Loading and releasing of drug in core–shell microgel systems [134].

of electrostatic interactions). The releasing of doxorubicin can also be controlled by adjusting temperature. Wu and his coworkers [134] have synthesized bimetallic (Ag-Au) NPs core and poly(ethylene glycol) P (EG) shell with hyaluronic acid surface. They also used these hybrid systems for anti-cancer treatment by loading the anticancer drug in core–shell system as shown in Fig. 8. The drug loaded systems were injected into the body. Laser-light then fell on the systems when the drug loaded systems reached the target place. The Au nanoparticles converted these lights into heat which increased the temperature of medium. Therefore, core–shell systems shifted from swelling state to deswelling state and drug come out from the sieves of core–shell systems along with water molecules. Wu et al [135] and Kim et al [136] have synthesized hybrid core–shell systems and then used for cancer treatment by loading doxorubicin.

6.1.2. Antibacterial activity

Bacteria are present in each place of environment and their attachment causes diseases in the human being. Therefore, it is very essential to kill these bacteria from our bodies. For this purpose, both core–shell microgel systems and hybrid core–shell microgel systems can be used for killing the bacteria. During the killing these bacteria, oxidation–reduction reactions take place which causes the killing of bacteria [137]. The structure of core–shell microgels also takes part in these oxidation–reduction reactions. Therefore, these bacteria can be killed by core–shell system [138]. For the purpose of killing the bacteria, the structure of core–shell systems is very important because these structures effect on the rapid oxidation–reduction reactions. Therefore, the effect of killing of bacteria by core–shell systems depends upon the structure of core–shell systems [139,140]. On the other hand, hybrid core–shell systems are more effective in killing the bacteria. In hybrid systems, some amount of metal nanoparticles escapes from the sieves of crosslinked polymeric network and starts to kill the bacteria along with core–shell microgel systems [141]. In this way, both metal nanoparticles and core–shell systems of hybrid core–shell microgels kill the bacteria. Therefore, the antibacterial activity of hybrid core–shell microgel systems is greater than core–shell microgels.

6.2. Catalysis

The core–shell microgels are not suitable as catalyst for catalytic reduction reactions. These core–shell microgels can be used as catalysts after the introductions of metal nanoparticles to form hybrid core–shell systems [27]. The catalytic performance of these hybrid core–shell systems depends on the nature of metal nanoparticles [48,49,142] and their position in core–shell morphology [38]. The hybrid core–shell in which metal nanoparticles are present in the core region shows the lowest catalytic performance due to the large distance between the surface of metal nanoparticles and reactants [143,144]. The hybrid core–shell systems, in which a layer of metal nanoparticles is present between the core and shell regions, exhibit more catalytic activity than the aforementioned hybrid systems due to less distance [64]. The hybrid core–shell systems in which metal nanoparticles are present in the shell region have more catalytic activity than previously discussed hybrid systems [38,74,145,146]. The hybrid systems in which the metal nanoparticles are present on the surface of crosslinked polymer to form shell have the highest catalytic activity than other hybrid core–shell systems due to smallest distance between the surface of nanoparticles and reactant molecules [57,147]. The catalytic activities of hybrid core shell systems can be further increased by conversing monometallic nanoparticles into bimetallic nanoparticles core–shell hybrid systems [58,63,148–153] as reported in our previous review article [154].

The catalytic performance of these hybrid core–shell systems can be tuned with the help of temperature [155], pH [156,157], ionic strength, and medium. These parameters affect the swelling and deswelling behavior of hybrid core–shell systems. Therefore, the diffusion rate of reactant molecules can be controlled by swelling/deswelling behavior as shown in Fig. 9. Diffusion rate is high at swelling state and low in deswelling state. The core–shell systems are in swelling state at $T \leq$ VPTT. Therefore, the catalytic activity is more at low temperature than high till a specific level. After that level of temperature, the catalytic activity of hybrid systems increases continuously with increasing temperatures due to increasing kinetic energy of reactant molecules.

Similarly, pH of medium also affects catalytic performance due to swelling and deswelling behavior. But the swelling and deswelling behavior of acidic and basic core–shell hybrid microgel systems are opposite to each other as discussed in sec 5.2. Therefore, their catalytic efficiency is also opposite with respect to each other.

6.3. As adsorbent

The core–shell microgels can be used as adsorbent for removal of pollutants from water as shown in Fig. 10 [14,47,158]. The structure of microgels have polar parts. These polar parts interact with the pollutants by ion–dipole interactions or dipole–dipole interactions or hydrogen bonding [46,159,160]. Therefore, the pollutants can easily be removed from water by core–shell microgels. the adsorption capacity and selectivity of core–shell systems can be enhanced by introducing the acidic [47,161] or basic [162] components in the structure of core–shell systems. These components enhance the adsorption capacity of core–shell systems due to swelling behavior and selectivity due to the formation of positive or negative charged structures of core–shell systems. These charged structures of core–shell systems attract the oppositely charged pollutants and repel the same charged pollutants. Therefore, selective adsorption of pollutants is occurred by acidic and basic microgels at different pH of the medium [29,163,164].

The adsorption of toxic metal ions can also be converted into metal nanoparticles with the help of a reducing agent such as $NaBH_4$ [46]. These synthesized nanoparticles also act as catalyst for reduction of other pollutants from water. Microgels are the best source for adsorption of metal ions due to this dual property (adsorbent and catalyst after conversion of metal ions into metal nanoparticles). Such behavior is not present in other systems. It is a new technique to remove the metal ions by adsorption as well as catalytic reduction.

6.4. For dye-sensitized solar cells systems

The structures of core–shell systems consist of polar parts which have partial positive and partial negative charges in their structure. The acidic and basic core–shell systems have the ability to donate and accept the protons respectively. In this way, the structure of core–shell systems is converted into ionic forms [165,166]. Due to these ionic forms, such types of core–shell systems are good conductors and therefore, they can be applicable in conducting materials. The structure of core–shell systems has such groups which have the capacity to donate electrons [167,168]. In this way, they can be applicable as auxochromes and decrease the energy gap present between the highest occupied molecular orbital to the lowest unoccupied molecular orbitals as shown in Fig. 11. In this way, the core–shell systems also increase the capacity of the materials which are used for dye-sensitized solar systems [169].

7. Conclusion and future directions

The core–shell systems consist of crosslinked organic polymer along with other organic or inorganic components. The combination of these materials is very interesting due to dual properties. This review article provides all classes of core–shell microgels on the basis of morphologies along with their suitable applications. The synthetic methods, which are used for required morphology of core–shell systems are discussed with proper orientation. The morphologies of core–shell systems can be identified with various techniques. These are important for justifications of proper designing of core–shell systems. These core–shell systems can be applied in various fields such as anticancer, antibacterials, adsorbent

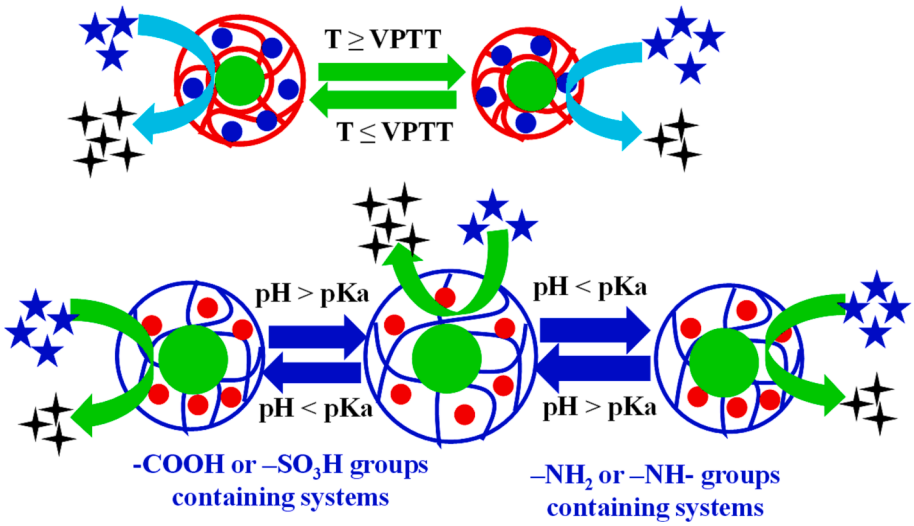

Fig. 9. Pictorial diagram of pH and temperature effect on catalytic efficiency of core–shell hybrid systems.

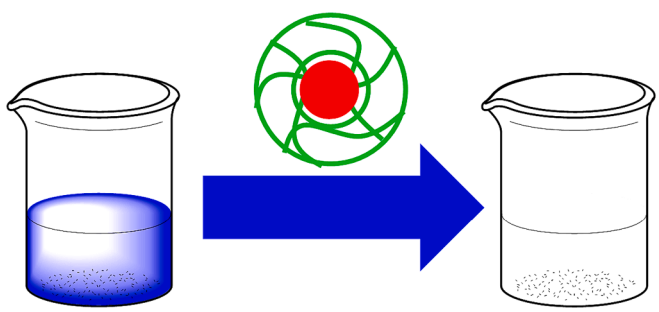

Fig. 10. Adsorption of pollutants from water by core–shell microgel systems.

for different pollutants from water, in dye-sensitized solar cell system, and catalysis. The core–shell systems consist of two different materials. In this way, these systems can be applied with respect to each component of core–shell systems. These components support each other due to synergistic effects and make suitable for various applications. These applications can be controlled by external environments of core–shell systems. The stimuli responsive behavior of core–shell systems can control their effect on their corresponding applications because the size of core–shell systems can be controlled by external stimuli such as pH, temperature, glucose, and ionic strength.

Core-shell systems serve as carriers for drug delivery purposes. Most drug delivery systems documented in literature rely on *N*-isopropylacrylamide and acrylic acid derivatives, which are thermo-responsive compounds that pose challenges in biodegradation. Substituting them with a biodegradable system in future experiments is advisable. These systems are generally underutilized for drug delivery. Consequently, both the previously reported systems and novel ones hold promise for drug delivery applications in upcoming research.

Metal nanoparticles integrated into core–shell systems have functioned as catalysts for a limited set of organic reactions, typically facilitating the reduction of 4-nitrophenol. Further exploration should examine the potential of different metal nanoparticles within core–shell systems to catalyze a wider array of organic transformations. Enhanced catalytic performance in reduction or degradation reactions has been observed in microgel systems loaded with bimetallic nanoparticles compared to those loaded with monometallic nanoparticles. Thus, extensive investigation into bimetallic nanoparticles loaded within core–shell microgel systems is warranted. In the near future, bimetallic nanoparticles containing hybrid systems can be investigated with stimuli responsive behavior such as pH, temperature, solvents and ionic strength. Further research is needed to assess the compatibility of substrate and scope, ensuring the functionality of core–shell systems. While the adsorption process has been employed in certain core–shell microgel systems. There is a potential for other core–shell systems to target the elimination of additional water contaminants, such as toxic organic dyes and heavy metal ions. These core–shell systems hold promise for purifying industrial waste through both adsorption and catalytic reactions.

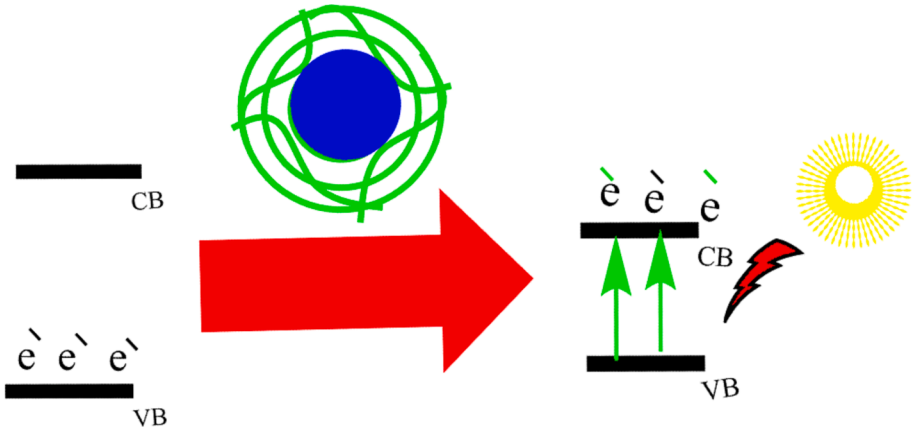

Fig. 11. Pictorial diagram for decreasing of energy gap between VB (valance band) and CB (conduction band).

However, the prominent challenge lies in the non-biodegradable nature of microgels in this context, a concern that could be addressed by adopting biodegradable monomers during the microgel synthesis process.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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